

Retail Sales Performance Analysis

Identifying Key Revenue Drivers and Seasonal Patterns

Portfolio Project – Data Analysis

Interactive dashboard, exploratory analysis, and actionable business recommendations

Simulating a real-world retail case using publicly available sales data to uncover growth opportunities.

Executive Summary

This project analyzes retail sales data to highlight the main drivers of revenue across stores and departments. The goal is to deliver actionable insights that support data-driven business decisions.

Key Takeaways

- Promotions have limited impact and should not be the main growth lever.
- Store segmentation by size is critical to maximize revenue.
- Focusing on high-performing departments (universal and specialized) drives most sales.
- Year-end seasonality affects large and medium stores; plan strategies accordingly.

Business Recommendations

- Promotions: Use tactically (inventory clearance, traffic boosts) but not as primary growth strategy.
- Store Segmentation: Adapt commercial strategies based on store size.
- Departments: Prioritize universal and specialized departments.
- Timing: Align marketing and operations with seasonal trends.

1. Data and Scope

Dataset

- Source: Kaggle – Retail Dataset
- Granularity: weekly sales by store and department
- Scope: multiple store types over several years

Limitations

- No margin or profitability data available
- No inventory or stock information
- Departments are anonymized

2. Store-Level Analysis

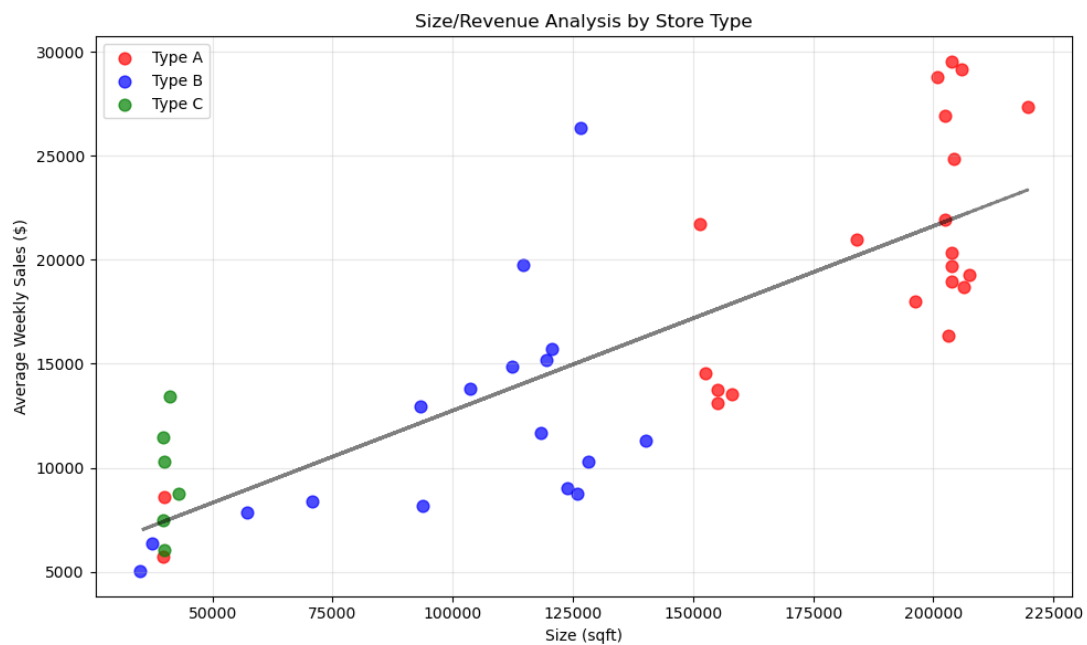
Key Insight

Revenue differences between stores are primarily driven by structural factors such as store size and location, not short-term operational decisions.

Implications for Business

- Prioritize investments and strategies by store size and type.
- Large stores (Type A) contribute most to overall revenue.
- Medium and small stores require targeted approaches to maximize returns.

Supporting Data



3. Promotion Impact Analysis

Key Insight

Promotions increase average revenue by only +1.9%, a marginal impact compared to structural factors like store size.

Implications for Business

- Short-term promotions alone are unlikely to drive significant growth.
- Focus on store-level levers (size, location, department mix) for stronger revenue impact.
- Use promotions tactically (inventory clearance, foot traffic), not as a primary revenue strategy.

Revenue Comparison

Promotions	Average Revenue	Increase (%)
Yes	16,177	+1.9
No	15,871	

4. Department-Level Analysis

Key Insight

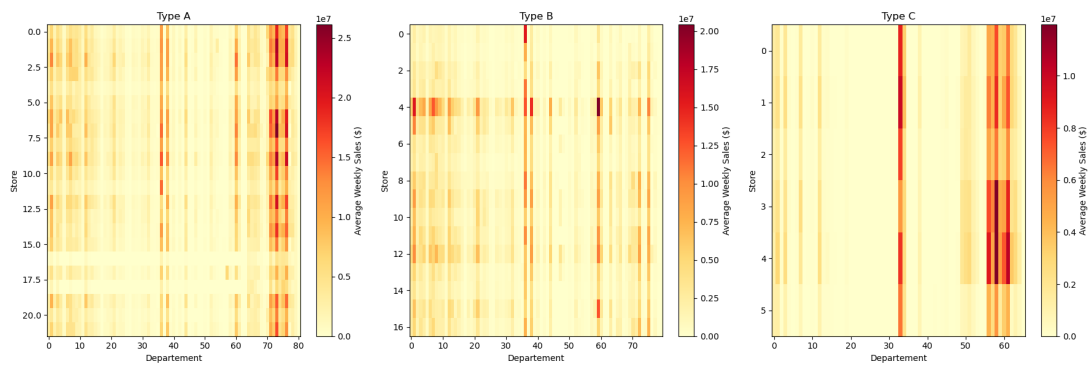
A few departments generate most of the store revenue.

Business Recommendations

- Prioritize universal and specialized departments to maximize revenue.
- Align inventory and promotions with high-performing departments.
- Allocate fewer resources to low-impact departments.

Supporting Data

Categories	Num. of Dept.	Avg. Presence (%)	Avg. Sales by Store
Universel	62	99.5	2,376,074
Spécialisé	4	13.9	1,698,715
Sélectif	2	50.0	247,287
Courant	13	83.1	207,299



5. Temporal and Seasonality Analysis

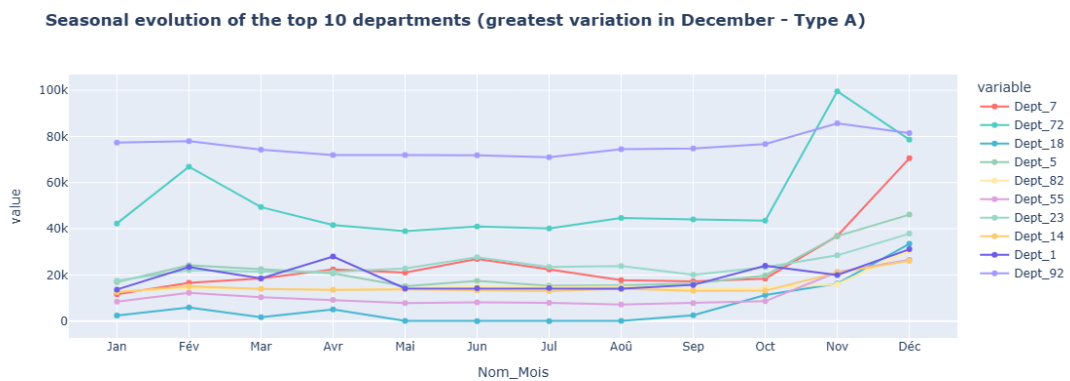
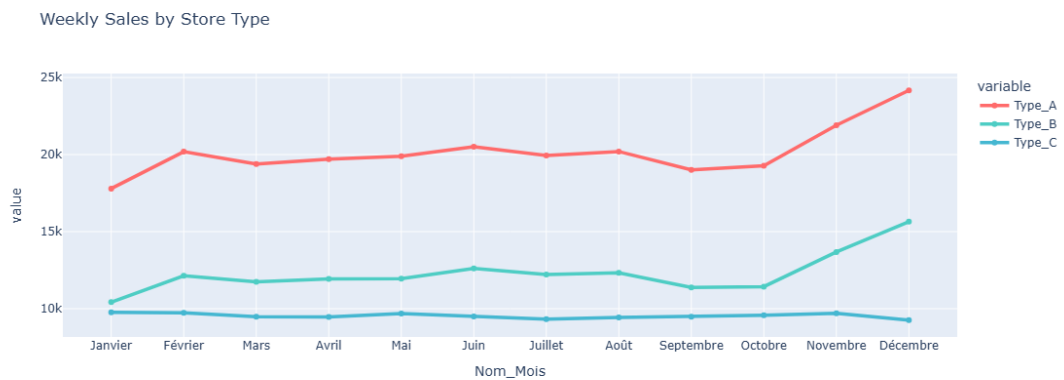
Key Insight

Year-end sales peaks mainly occur in large and medium stores, driven by a few high-performing departments.

Business Recommendations

- Align inventory and staffing with year-end peak periods.
- Focus marketing on departments driving peak sales.
- Small stores maintain steady sales and need less seasonal adjustment.

Supporting Visuals



6. Potential Extensions

Additional data could enable the following business opportunities:

- Analyze profitability and margins to identify high-return products and stores.
- Optimize promotion strategies to maximize ROI.
- Improve inventory planning and reduce stock-out risk.

Deliverables

- [Interactive Streamlit dashboard \(FRENCH\)](#)
- [Jupyter notebook / methodology \(FRENCH\)](#)
- [GitHub](#)